

# NEURAL ODE FOR ALZHEMER'S DISEASE

Recent Progress

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Yilin Li

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The main reason the previous results did not show an S-shaped curve was excessive data noise: after preprocessing, very little data remained, and the distribution of this data made it almost impossible to induce an S-shaped trajectory from the model.

Adjusted the data preprocessing pipeline:

1. Allowing time points with partially missing data to be included in the training set (meaning the number of sample points for each biomarker can differ). The data distribution after this processing is closer to that shown in the original paper's figures.
2. Since the model only needs one complete initial value to auto-regressively predict the entire trajectory, we don't need to worry about incomplete data at some time points. We can simply ignore these points when calculating the loss.

Neural networks have such strong learning and expressive capabilities that they are harder to train into an S-shape than polynomial ODEs. To overcome the influence of noise and encourage the model to show an S-shape, we use the following pipeline:

1. Fit four Sigmoid functions to the four biomarkers respectively
2. Analytically compute the derivatives of these Sigmoid functions on a grid
3. Train the neural network to approximate these derivatives at each grid point

The step 1 has a high degree of randomness. The learned Sigmoid curves can vary greatly each time, leading to vastly different Neural ODE performances. We made multiple attempts and selected the few results that looked best visually to display.

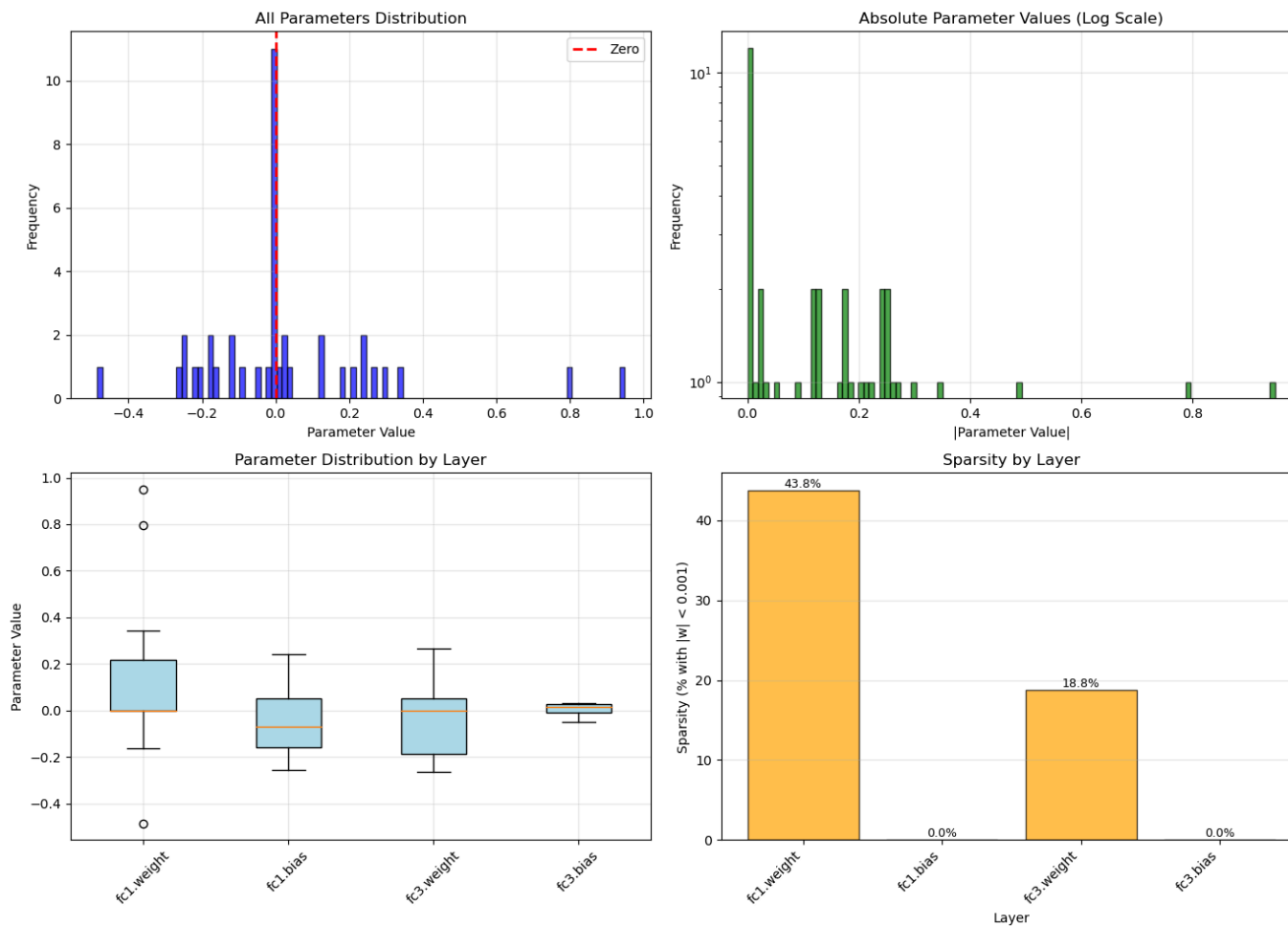
In particular, practice has shown that the smaller the model size, the better (currently using a single hidden layer of width 4). A slightly larger model (e.g., a single hidden layer of width 16 or more) is extremely prone to overfitting during the main training phase.

We specifically studied how to reduce the model size. We applied L1 regularization to encourage sparsity. Then, we counted the number of parameters close to 0 (absolute value  $< 0.001$ ).

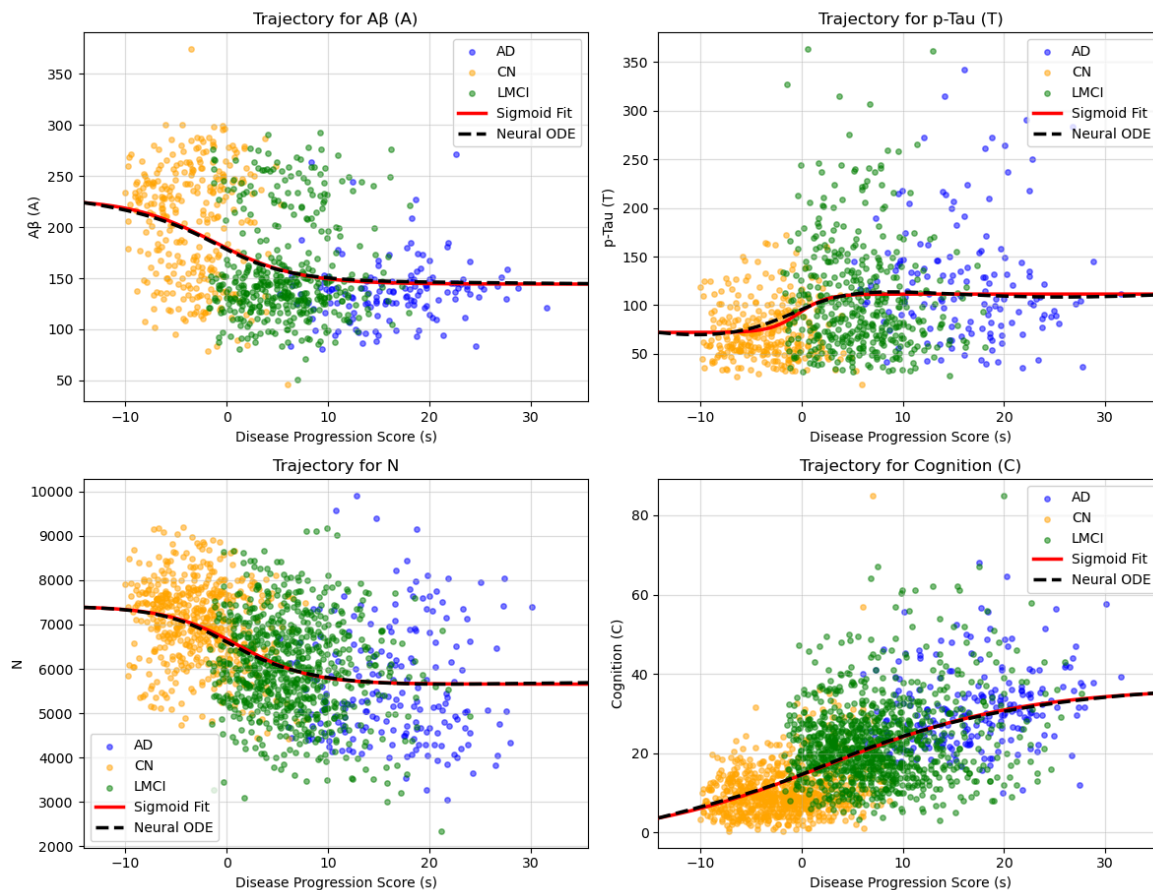
In more complex models, this proportion was often above 90%. By reducing the model size, this ratio has been brought down to below 38%. The inactive parameter ratio for the final adopted model was 37.5%.

Neural Network Parameter Statistics:

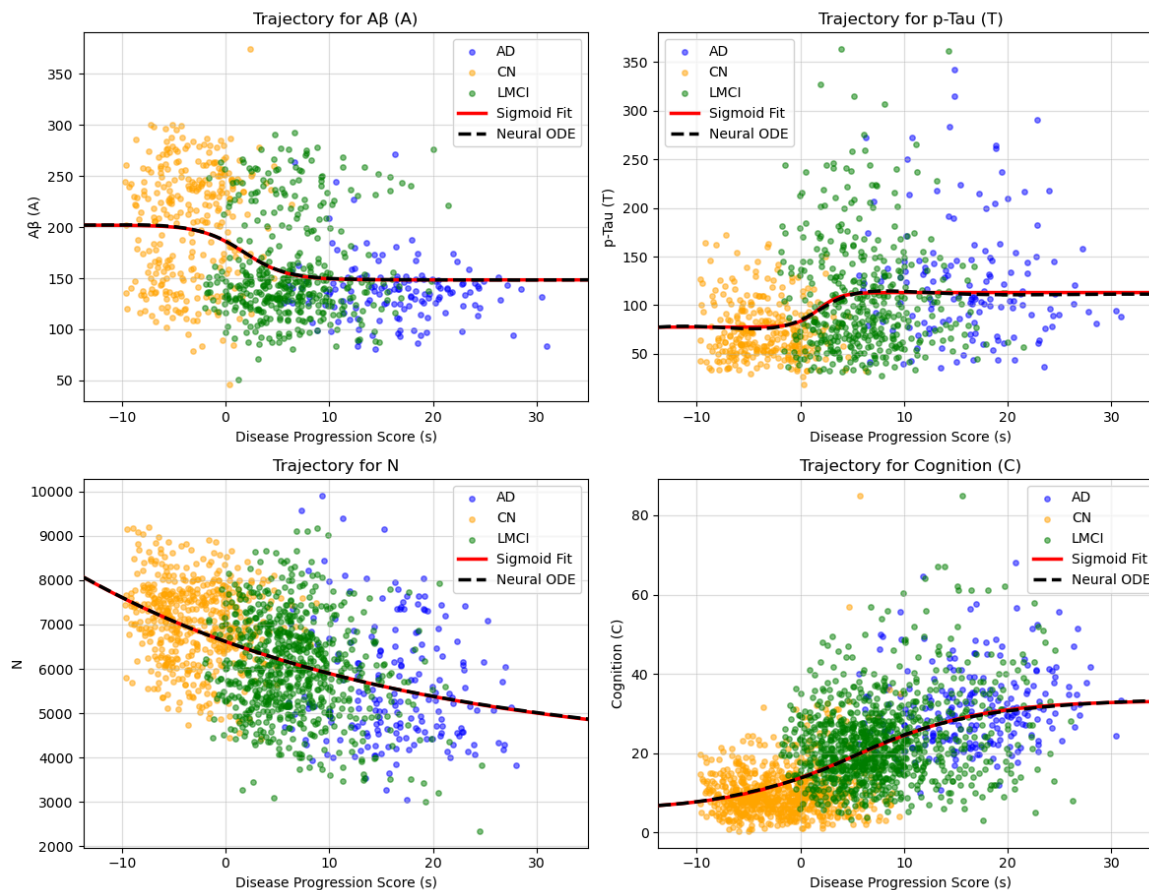
|                                         |           |
|-----------------------------------------|-----------|
| Total parameters                        | 40        |
| Mean                                    | -0.041059 |
| Standard Deviation                      | 0.291287  |
| Min                                     | -0.695715 |
| Max                                     | 0.704539  |
| Proportion close to 0 ( $ w  < 0.001$ ) | 37.50%    |



AD Biomarker Trajectories: Sigmoid Fit vs. Neural ODE

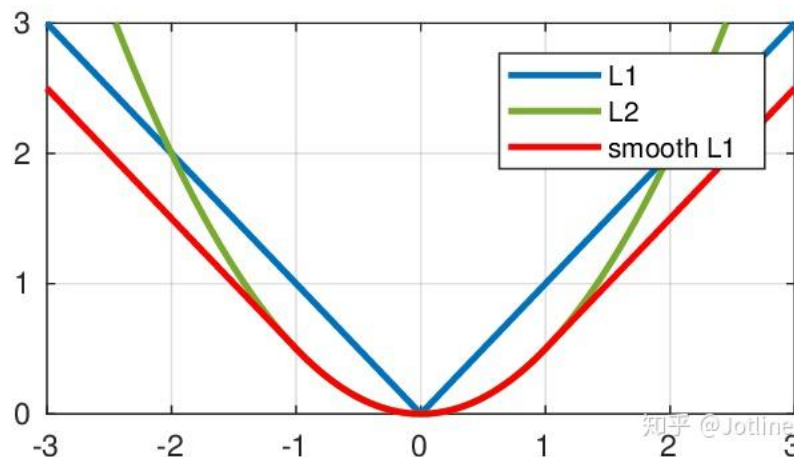


AD Biomarker Trajectories: Sigmoid Fit vs. Neural ODE

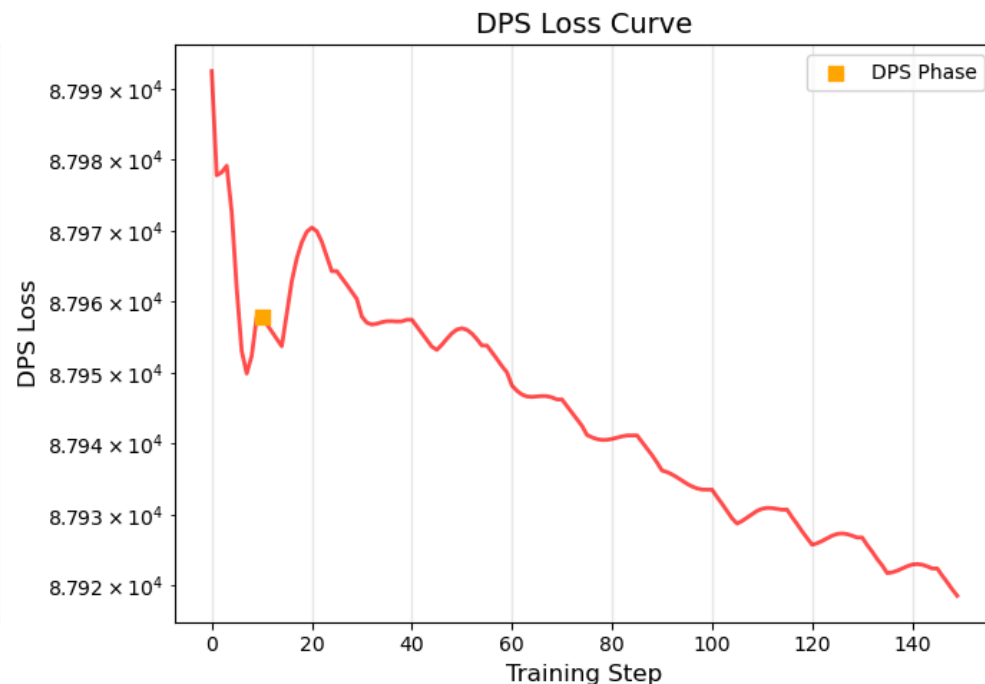
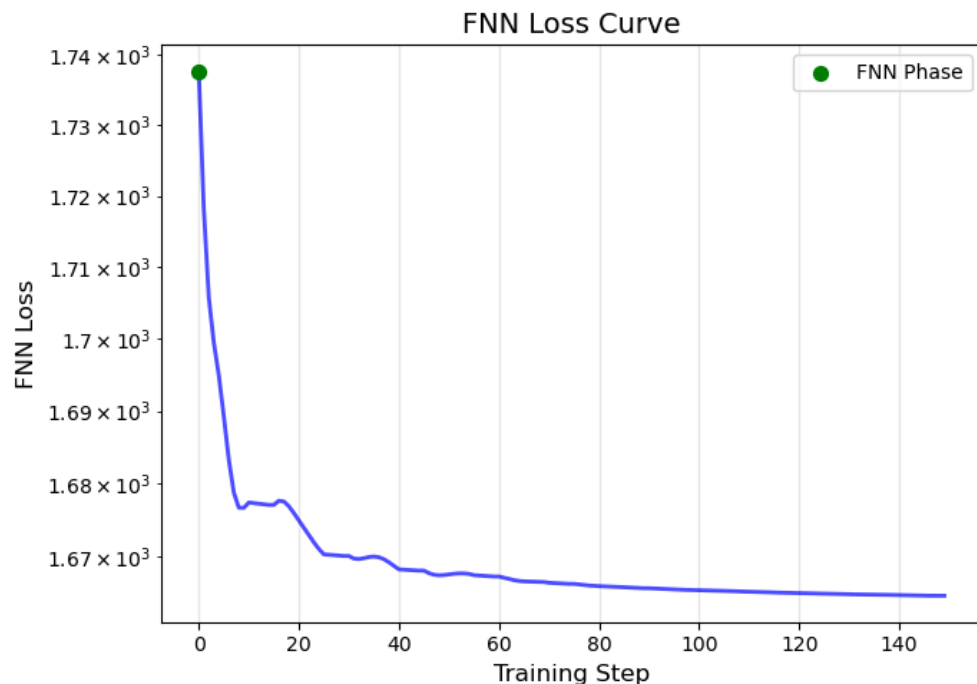


The pretrained model is then calibrated on the actual data. We use SmoothL1Loss instead of L2 Loss, as the former is less sensitive to outliers. L1 regularization is still applied to the model.

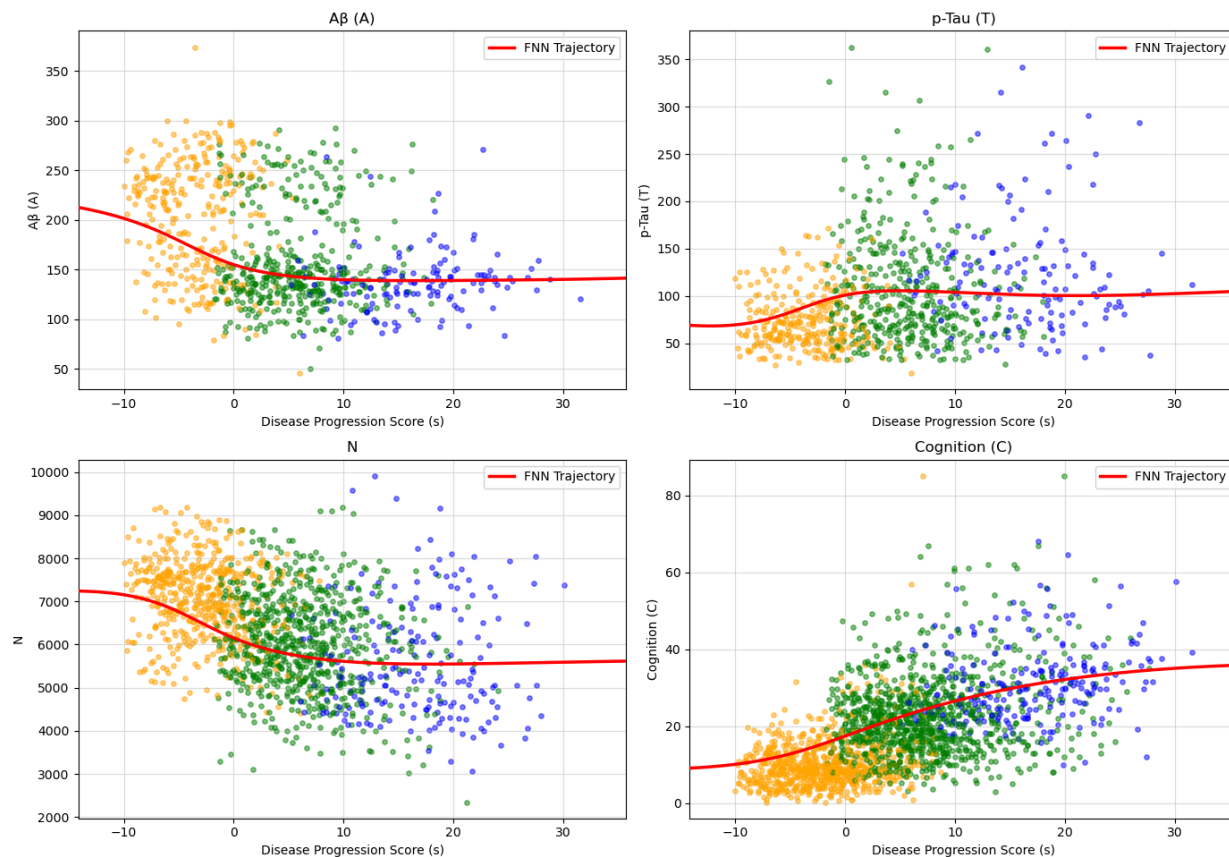
$$\text{loss}(x, y) = \frac{1}{n} \sum_{i=1}^n \begin{cases} 0.5(y_i - f(x_i))^2, & \text{if } |y_i - f(x_i)| < 1 \\ |y_i - f(x_i)| - 0.5, & \text{otherwise} \end{cases}$$



Alternating Training: In each round, we first optimize the model parameters for 10 iterations, then optimize the dps parameters for 5 iterations. The faster Adam algorithm is used for both parts. We run this for a total of 10 Epochs.



FNN Model with Alternating Optimization (Adam)



FNN Model with Alternating Optimization (Adam)

